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INVOIC GUARD

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

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1. Introduction

1.1 Project Name

Invoice Guard

1.2 Purpose

Invoice Guard is a web-based invoice verification and fraud detection system designed to help businesses identify suspicious duplication, manipulated, or invalid invoices before processing payment.

1.3 Scope

The system allow user to upload invoices, validate invoice information, detect potential fraud indicators, generate risk assessments, and maintain a secure audit trail of invoice verification activities.

1.4 Objectives

- Detect duplicate invoices.
- Identify suspicious invoice patterns.
- Verify invoice details.
- Generate fraud risk scores.
- Maintain invoice verification history.
- Reduce financial losses caused by invoice fraud.

2. Stakeholders

Primary User

- Finance Officers
- Accountants
- Auditors
- Business Owners

Secondary Users

- System Administrators

3. Functional Requirements

FR-01 User Authentication

- Users shall be able to register and log in.
- Users shall be able to reset passwords.

FR-02 Invoice Upload

- User shall be able to upload invoices to the system.
- The System shall support PDF, JPG, JPEG and PNG formats.

3. Functional Requirements

FR-02 Invoice Upload

- Users shall be able to upload invoices to the system.
- The system shall support PDF, JPG, JPEG, and PNG formats.
- The system shall reject unsupported file formats.

FR-03 QR Code Scanning

- Users shall be able to scan a QR code printed on an invoice.
- The system shall extract the information encoded in the QR code.
- The system shall verify whether the QR code is valid.
- The system shall identify missing, invalid, or unreadable QR codes.

FR-04 Invoice Data Extraction

- The system shall extract relevant information from uploaded invoices.
- The system shall extract information such as invoice number, supplier, date, subtotal, tax, and total amount.
- The system shall identify when required information cannot be extracted.

FR-05 Invoice Information Validation

- The system shall validate the extracted invoice information.
- The system shall verify invoice numbers, dates, supplier details, and monetary values.
- The system shall compare information from different sources for inconsistencies.

FR-06 Duplicate Invoice Detection

- The system shall detect duplicate invoices.
- The system shall compare invoice numbers, supplier details, dates, and amounts to identify possible duplicates.
- The system shall notify users when a possible duplicate is detected.

FR-07 Invoice Tampering Detection

- The system shall compare information extracted from the invoice with information contained in the QR code.
- The system shall identify inconsistencies between the invoice and QR code data.
- The system shall flag invoices with suspected modifications.

FR-08 Fraud Risk Analysis

- The system shall analyze invoices for predefined fraud indicators.
- The system shall calculate a risk score for each invoice.
- The system shall classify invoices according to their risk level.

3. Functional Requirements

FR-09 Verification Result

- The system shall display the verification result to the user.
- The result shall indicate whether the invoice is verified, suspicious, or invalid.
- The system shall display the reasons for a suspicious or invalid result

FR-10 Alerts and Notification

- The system shall notify users when suspicious activity is detected.
- The system shall identify the specific reason for the alert.

FR-11 Invoice History

- Users shall be able to view previously verified invoices.
- The system shall store the verification status and results of each invoice.

FR-12 Reports

- Users shall be able to generate invoice verification reports.
- Reports shall include invoice details, verification results, detected issues, and risk scores. .

FR-13 Audit Logging

- The system shall record invoice verification activities.
- The system shall record the user, date, time, and verification result.
- Audit records shall not be editable by normal users.

FR-14 User Management

- Administrators shall be able to create, update, and deactivate user accounts.
- Administrators shall be able to assign user roles and permissions.

4. Non-Functional Requirements

Performance

- Invoice analysis shall complete within 10 seconds.

Security

- All user passwords shall be encrypted.
- Data transmission shall use HTTPS.

Reliability

- Invoice analysis shall complete within 10 seconds.

4. Non-Functional Requirements

Usability

- Users shall be able to complete invoice verification with minimal training

Scalability

- The system shall support future intergration with ERP and accounting system

5. System Constraints

- Internet connection required.
- Supported formats: PDF, JPG, PNG.
- User authentication required for system access.

6. Assumptions

- Uploaded invoices contain readable text.
- Users provide accurate invoice information.
- Supplier records exist in the database when verification is performed.

7. Success Criteria

- Detect duplicate invoices accurately.
- Reduce manual verification effort.
- Generate fraud detection reports.
- Maintain complete audit records.

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